AI-Powered Equity Research Agent

Finance 372T/397 · Group Project Pitch · April 7, 2026

THE PROBLEM

- Fundamental data, filings, and sentiment signals are scattered across terminals and databases
- Free tools don't easily combine accounting/profitability signals with textual/sentiment analysis in one view
- Result: investors miss the divergence between what the numbers say and what management says

THE SOLUTION

- Input any U.S. equity ticker →
  - Quant dashboard: Piotroski F-Score, gross profitability, earnings quality, valuation & momentum
  - Sentiment & textual analysis layer: scores earnings call transcripts for tone, hedging, and confidence — tracked quarter-over-quarter
  - AI synthesis: Claude reconciles quant signals vs. sentiment signals and flags divergences, producing a plain-English bull case, bear case, and key risks

DASHBOARD MODULES

1. **Piotroski F-Score**
   9-point financial strength score · component breakdown
2. **Gross Profitability**
   Novy-Marx (2013) signal · percentile rank vs. sector
3. **Earnings Quality**
   Accruals ratio · cash flow vs. net income divergence
4. **Valuation & Momentum**
   EV/EBITDA · P/E · 12-1 month momentum · reversal flags
5. **Earnings Call Sentiment & Textual Analysis**
   Tone score · hedging language · QoQ confidence trend
6. **AI Synthesis**
   Claude reconciles quant + sentiment → bull/bear verdict
   - *Financials look strong, but management tone has deteriorated three quarters in a row — flag for further review*

Execution Plan

WEEK 1
April 7 – 11

Data & Metrics Engine

- Pull Compustat fundamentals: (income, balance sheet, cash flow) for all U.S. equities
- Build signal calculators: Piotroski F-Score, gross profitability, accruals ratio
- Add CRSP market data: returns, momentum, valuation multiples
- Compute percentile ranks within sector & universe

WEEK 2
April 12 – 16

AI Layer & Dashboard

- Fetch earnings call transcripts via SEC EDGAR; build sentiment scoring pipeline
- Score tone, hedging language, forward-looking confidence; track QoQ trend
- Claude API: interpret quant + sentiment signals → bull case, bear case, key risks
- Build Streamlit dashboard: ticker input → visual output; test across cap ranges

WEEK 3
April 17 – 21

Validation & Polish

- Test dashboard on tickers across cap ranges; confirm outputs are coherent and actionable
- Refine Claude prompts based on output quality; fix any data or display issues
- Finalize live demo and presentation slides before April 21
- Incorporate feedback; final writeup due April 27

Tools: Python · WRDS (Compustat, CRSP) · Claude API · SEC EDGAR · Streamlit

Division of Labor

Blake Stanley
Data Engineering & Quant Analysis

RESPONSIBILITIES

- WRDS data pipeline: Compustat fundamentals + CRSP market data
- Implement signal calculators: Piotroski F-Score, gross profitability, accruals, momentum
- Composite scoring engine & percentile rankings vs. sector/universe
- Polishing, presentation slides, and final writeup (working together)

Deliverable: Working metrics engine by April 11

Will Pechersky
AI Layer, Sentiment & Dashboard

RESPONSIBILITIES

- Earnings call transcript fetching & sentiment scoring pipeline (tone, hedging, QoQ trend)
- Claude API integration: prompt engineering for quant + sentiment synthesis
- Streamlit dashboard: ticker input → visual output; end-to-end testing
- Polishing, presentation slides, and final writeup (working together)

Deliverable: Working dashboard by April 16